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Adaptive surrogate-based design optimization with expected improvement used as infill criterion

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A prominent advantage of using surrogate models in structural design optimization is that computational effort can be greatly reduced without significantly compromising model accuracy. The essential goal is to perform the design optimization with fewer evaluations of the typically finite element analysis and ensuring accuracy of the optimization results. An adaptive surrogate based design optimization framework is proposed, in which Latin hypercube sampling and Kriging are used to build surrogate models. Accuracy of the models is improved adaptively using an infill criterion called expected improvement (EI). It is the anticipated improvement that an interpolation point will lead to the current surrogate models. The point that will lead to the maximum EI is searched and used as infill points at each iteration. For constrained optimization problems, the surrogate of constraint is also utilized to form a constrained EI as the corresponding infill criterion. Computational trials on mathematical test functions and on a three-dimensional aircraft wing model are carried out to test the feasibility of this method. Compared with the traditional surrogate base design optimization and direct optimization methods, this method can find the optimum design with fewer evaluations of the original system model and maintain good accuracy.

Keywords: design optimization; surrogate modelling; adaptive surrogate based design optimization; expected improvement

AMS Subject Classifications: 65K10; 90C06; 90C59; 42A61

1. Introduction and state of the art

In structural design problems, system responses are typically acquired through the finite-element method (FEM), which can be computationally expensive. The optimization of such models is a demanding problem, which requires simulation on a number of design variables in a large design space. Traditional direct optimization (DO) may be excessively time-consuming using either sequential quadratic programming (SQP) or genetic algorithm (GA). To accelerate the optimization process, investigations are made on surrogate models, which are

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actually a combination of simple functions that require significantly less computational effort to approximate the system response. Compared with the true system model, surrogate models permit us to explore the system more efficiently. Surrogate models have been extensively studied and applied to engineering design field, for example, Forsberg [6] and Blumhardt [2] used surrogate model to assist the structural optimization of vehicle crash simulation problems, Booker [3] developed effective numerical methods for managing the use of surrogate model in design optimization of a helicopter. Huang [7] applied surrogate model to the shape optimization of an aero-engine turbine disc. Furthermore, there are many other examples of the application of surrogate models in computer aided engineering and computer aided design described by Kleijnen [8] and Simpson [16].

To build a surrogate model, there are basically two steps: (1) design of experiment (DoE) or sampling, (2) surrogate modelling or metamodelling, which means after getting properly distributed samples on the design space, surrogate models are built with interpolation or regression algorithms. There are a number of DoE samplings and surrogate modelling methods available. For sampling, there are Latin hypercube sample (LHS), orthogonal arrays [11], Hammersley sequence, uniform designs, etc. For approximation, there are Kriging, polynomial regression, radial basis function, support vector regression [19], artificial neural networks, etc. The choice of such methods can greatly affect the quality of the surrogate model, and thus should match the case being studied, the available sample data and performance requirements. Zhao and Xue [21] studied different characteristics of sample data working with four types of surrogate modelling methods considering accuracy, confidence, robustness and efficiency. Several articles are published in engineering design field to test and compare the performance of different surrogate modelling methods. For example, Simpson [18] compared the performance of a second-order response surface model and Kriging model based on the multidisciplinary design problem of an aero-spike nozzle. He suggested Kriging as an alternative technique with potential improvements to the other models. Simpson [17] also contrasted five sampling methods and four surrogate models in terms of their capability to generate accurate approximations for two engineering applications. Results revealed that Kriging provides accurate approximations for a variety of sampling methods and sample sizes. Paiva [12] studied different cases based on aircraft wing design problems to compare different surrogate models. He concluded that for higher dimensional problems, the usage of Kriging is more efficient.

In this article, LHS and Kriging are chosen to build surrogate models. Besides the choice of sampling and modelling methods, the measure of quality for surrogate models is also very important. Details of LHS, Kriging and the quality measures of surrogate models are introduced in Section 2. A comprehensive review of different DoEs, surrogate modelling methods and optimization frameworks, including thorough implementation results can also be found in the work of Quiepo et al. [13].

To assist the sampling and surrogate modelling methods, a preliminary study of the inputs and outputs relationship of the system model can be very beneficial. This helps to decide which parameters are to be sampled and to be approximated. An example of this tactic is given in Section 3.

The focus of this article is on the framework of surrogate based design optimization (SBDO). Traditionally, a sequential approach with only a one-shot, space filling DoE is used. The problem is that we have no *a priori* knowledge of the

efficiency and accuracy of the surrogate model based on such samples. To cope with the shortcomings, an adaptive approach refining surrogate models iteratively based on the feedback information of the optimization results is proposed. Adaptive strategies in the framework of SBDO have also been studied. For example, Li [9] used accumulative error as infill criterion of the adaptive DoE approach to assist building a global accurate surrogate model, following the principles of a greedy algorithm. However, in his study, the adaptive activity is not related to the optimization process, but instead, only within the surrogate modelling process. Couckuyt [4] used expected improvement EI as an infill criterion for adaptive surrogate based optimization. Although this was not tested with engineering problems, it showed potential of improving the efficiency of optimization.

In this article, EI is used as the criterion, which is calculated with the information fed back by the optimization results of each iteration. An adaptive framework considering the whole optimization process is designed and tested with both mathematical test examples and a three-dimensional airplane wing model. For constrained optimization problems, an infill criterion based on constrained EI (CEI) is used. Descriptions of the framework and the infill criteria are provided in Section 4. Tests and comparisons of different approaches are given in Section 5. This is followed by conclusion in Section 6.

2. DoEs and surrogate modelling

In this section, LHS and Kriging are introduced, together with the reason why they are chosen as DoE and surrogate modelling methods in this research. The quality measures of the surrogate models, especially for engineering problems are also discussed.

2.1. *Latin hypercube sampling*

LHS is an extension of quota sampling [20], and it is a statistical method to generate a distribution of parameters. This ensures that data points are well-scattered for the design space of any dimension. For a two-dimensional problem, the samples are distributed in a square grid with only one sample in each column and each row. For higher dimensional space, the solution is similar. Given the sample size M and dimension number N, the design space is cut into M parts in each dimension. If each sample point is located in a fixed position of a grid, then the maximum number of combinations of outputs equals $(M!)^{N-1}$. There can be more combinations if samples are located randomly in the grids. For example, to generate three sample points in a two-dimensional design space, there are basically six combinations, which can be varied by moving the point within each grid, as shown in Figure 1. One advantage of LHS is that the output is dominated by only a few of the components of the design space, and each of those components is represented in a fully stratified manner [10]. That is to say, it can easily generate well-distributed samples in any dimensional design space no matter whether the sample size is large or small. Each point is not necessarily located in the centre of the grid it belongs to, but has some random mutation within the grid. Such kind of randomness increases the diversity of LHS combinations, which can be used to test the stability of the surrogate

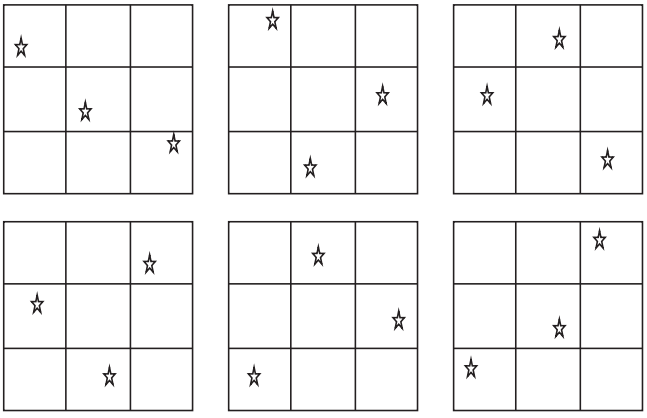

Figure 1. Combinations of the LHSs with three points in two-dimensional design space.

modelling methods. For situations where uniformity of sample distribution is needed, the original LHS can be adjusted by setting minimum distance or correlation requirements to the LHS method.

2.2. *Kriging*

The Kriging method was first developed by Danie Krige, a mining engineer, and made its way into engineering design following the work of Sacks [15]. He applied the method to the approximation of computer experiments [5]. Now Kriging is widely used as a Gaussian stochastic process-based approximation method to interpolate the value at an unobserved location from the observed information. Kriging model consists of two parts. One is a simple regression part to fit the samples according to the rule of best linear unbiased predictor [14]. The other is a correlation part to adjust the prediction error at the locations near the observed places by maximum likelihood estimation [1].

The Kriging model starts with the sample points obtained through a DoE, $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n\}^T$, and their respective responses $\boldsymbol{y} = \{y_1, y_2, \ldots, y_n\}^T$. $\mathbf{X}$ is an $n \times p$ matrix, where n is the number of sample points, and p stands for the dimension number of the design variables. If $\boldsymbol{x}$ is the point where a prediction value $\hat{y}$ is defined, a Kriging model is formulated as

$$\hat{y} = \sum_{i=0}^{n} \beta_i f_i(\boldsymbol{x}) + Z(\boldsymbol{x}), \tag{1}$$

where $\sum_{i=0}^{n} \beta_i f_i(\boldsymbol{x})$ is the regression model, $Z(\boldsymbol{x})$ is the correlation model, which is a Gaussian process with the mean value 0 and covariance σ given by

$$Cov(Z(x_j), Z(x_k)) = \sigma^2 R_{jk}(\theta, x_j, x_k), \tag{2}$$

R_{jk} is the Gaussian correlation function:

$$R_{jk}(\theta, x_j, x_k) = \prod_{i=1}^{p} e^{-\theta_i(|x_{ji} - x_{ki}|^2)}. \tag{3}$$

An interpolation model is then built, which assumes that no error is allowed on the observed data. By forming the likelihood function and defining

$$\psi = \begin{pmatrix} R(\boldsymbol{\theta}, x_1, \boldsymbol{x}) \\ \vdots \\ R(\boldsymbol{\theta}, x_n, \boldsymbol{x}) \end{pmatrix}, \quad \boldsymbol{\Psi} = \begin{pmatrix} R_{11} & \cdots & R_{1n} \\ \vdots & \ddots & \vdots \\ R_{n1} & \cdots & R_{nn} \end{pmatrix},$$

maximum likelihood estimation for $\boldsymbol{\mu}$ and σ^2 is obtained by

$$\hat{\boldsymbol{\mu}} = \frac{1^T \boldsymbol{\Psi}^{-1} \boldsymbol{y}}{1^T \boldsymbol{\Psi}^{-1} 1}, \quad \hat{\sigma}^2 = \frac{(\boldsymbol{y} - \mathbf{1}\boldsymbol{\mu})^T \boldsymbol{\Psi}^{-1} (\boldsymbol{y} - \mathbf{1}\boldsymbol{\mu})}{n}. \tag{4}$$

$\boldsymbol{\theta}$ is then determined by maximizing the likelihood function or minimizing the corresponding negative logarithmic function Neglog$L(\hat{\boldsymbol{\mu}}, \hat{\sigma}^2, \boldsymbol{\theta})$:

$$\boldsymbol{\theta} = \left\{ \boldsymbol{\theta} | \min \ NeglogL(\hat{\boldsymbol{\mu}}, \hat{\sigma}^2, \boldsymbol{\theta}) = \frac{n}{2} \ln(\hat{\sigma}^2(\boldsymbol{\theta})) + \frac{1}{2} \ln |\boldsymbol{\Psi}(\boldsymbol{\theta})| \right\}. \tag{5}$$

Finally, we can predict with a Kriging model at any position within the design space [5] with

$$\hat{\boldsymbol{y}}(\boldsymbol{x}) = \hat{\boldsymbol{\mu}} + \psi^T \boldsymbol{\Psi}^{-1} (\boldsymbol{y} - \mathbf{1}\hat{\boldsymbol{\mu}}). \tag{6}$$

As introduced above, to build a Kriging model, after getting the most suitable value of $\boldsymbol{\theta}$, we have to calculate the matrices including $\boldsymbol{\Psi}$, $\boldsymbol{\psi}$, $\hat{\boldsymbol{\mu}}$ and $\hat{\sigma}^2$. These matrices are also very important for making any modification to the basic Kriging. For example, the infill criterion, which is used to improve the accuracy of Kriging model by filling points to the sample sets, is derived based on those formulations. The infill criterion is called EI, and is introduced in Section 4.

2.3. *Posterior quality measures of surrogate models*

The indexes such as accuracy, efficiency, robustness and stability are usually used to measure the quality of surrogate models. In this work, we will mainly look into the accuracy and efficiency. Robustness is considered by repeating each approach 100 times and using statistic value to compare the performance of different methods.

In engineering optimization problems, we use the extreme values $\boldsymbol{x}^*$ and f^* to measure the accuracy of the optimization strategy. Extreme values are the final optimum results of an optimization problem. In the following tests, the extreme values calculated by each method are compared with the standard values or benchmark values $\boldsymbol{x}^*_{ben}$ and f^*_{ben} in order to measure the accuracy of each method.

Different optimization strategies may find different extreme values, local or global, feasible or unfeasible, depending on the type of the searching algorithm and the convergence condition. In this work, we define the acceptable extreme values as $\boldsymbol{x}^*_{acc}$ and f^*_{acc}. Supposing that we are minimizing the objective function, either of the following two equations should be satisfied for a design to be

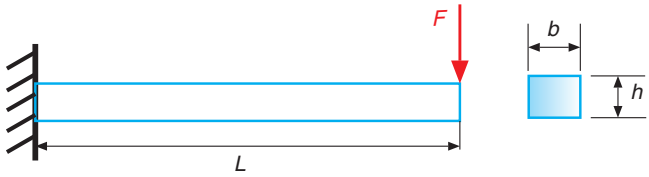

Figure 2. Cantilever beam model under vertical force on the free end.

considered acceptable:

$$\frac{\|\boldsymbol{x}^*_{acc} - \boldsymbol{x}^*_{ben}\|}{\|\boldsymbol{x}^*_{ben}\|} \leq \delta_x, \tag{7}$$

$$\frac{f^*_{acc} - f^*_{ben}}{f^*_{ben}} \leq \delta_f, \tag{8}$$

where δ_x and δ_f are predefined acceptable error for the design parameters and the objective function value, respectively. $\|\cdot\|$ stands for the Euclidean distance.

The percentage of the acceptable extreme values found by each method among the 100 repetitions is then p_{acc}:

$$p_{acc} = \frac{n_{acc}}{100}, \tag{9}$$

where n_{acc} is the number of acceptable extreme values found in the 100 repetitions. p_{acc} represents the chance of each method to find global optimum and to search for best solution with constraint conditions.

For assistance of accuracy measurement, the root mean square error (RMSE) of the optimum design variables $rmse(\boldsymbol{x}^*)$ and the optimum objective values $rmse(f^*)$ of the 100 runs are used.

The average number of evaluations $\overline{n}_{eva}$ of the true model needed for optimization is used to measure the efficiency of a method. This is because that the evaluation of the true model in engineering optimization is the most time-consuming part. In contrast, time expense for computing surrogate models is relatively small and sometimes can even be neglected. In this study, as the optimization process is run 100 times for each method, the averaged number of evaluations of the true model is used to measure efficiency.

3. Improving quality by choice of parameters to be surrogated

To improve the quality of surrogate models, there is one important aspect besides choosing the sampling and modelling methods: the study of the relationship between inputs and outputs of the system before deciding which parameters to be sampled and surrogated. The choice of variables to be sampled and approximated plays a large role in modelling accuracy and efficiency. A structural example is shown here to explain this concept. In Figure 2, the bending stress σ_b is the output of the model, the inputs are the design variables: the width b and height h of the cross-section.

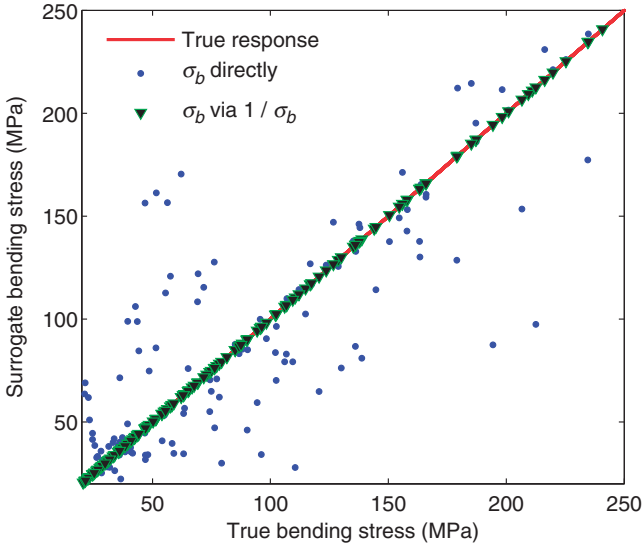

Figure 3. Surrogate models by direct and indirect surrogate modelling.

As we know, σ_b can be obtained by the following expression:

$$\sigma_b = \frac{M_b}{I_z} \cdot \frac{h}{2} = \frac{6FL}{bh^2}, \tag{10}$$

where σ_b is the bending stress, M_b is the bending moment, F is the force and I_z is the moment of inertia according to the z-axis, which is perpendicular to the beam plane. As we can see, σ_b is in inverse relationship with the inputs b and h. While the inverse σ_b, or $\frac{1}{\sigma_b}$, is direct proportional to the inputs, building surrogate model for $\frac{1}{\sigma_b}$ subject to b and h is more reasonable than building model for σ_b directly. In this example, LHS is used for sampling, Kriging is used for approximation. The same 25 sample points are generated for both methods. Comparison of surrogate models built with σ_b directly and indirectly is demonstrated in Figure 3.

In Figure 3, the horizontal and the vertical axes represent the true σ_b and the surrogate σ_b, respectively. An accurate surrogate model should be exactly in accordance with the diagonal line shown in the figure. We can see that building σ_b indirectly by approximating $\frac{1}{\sigma_b}$ first can obtain higher accuracy. Therefore it is important to preliminarily study the relationship between inputs and outputs, especially for structural models. Engineering experience of system parameters can be used to decide which of them to be sampled and surrogated. This strategy is used in the tests of the FEM model in Section 5.

4. Framework of SBDO

In this section, the traditional way of proceeding with SBDO is introduced, and its problems are discussed. Afterwards, an adaptive way of SBDO is proposed to overcome these problems. The frameworks of the two methods are given,

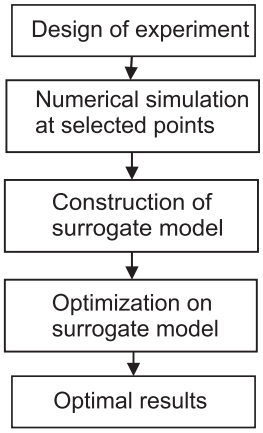

Figure 4. Framework of SSBDO.

respectively, to demonstrate their differences. The features of the adaptive method are further expounded, including the infill criterion and the stopping criterion. Finally, the implementation of the ASBDO method is given with more specific framework and methodical description.

4.1. *Sequential surrogate based design optimization*

The sequential SBDO (SSBDO) defined here is the traditional method for SBDO. There is no feedback from the surrogate model or the optimal results into the DoE process. Its framework is shown in Figure 4.

It can be seen in Figure 4 that SSBDO is carried out with only one-shot sampling, and then optimization is performed based on this surrogate model. The problem is that the best sample size, which plays an important role to the quality of the surrogate model, is not known *a priori*. With insufficient sample size, the approximation error for points far away from the samples can be quite large, and oscillation problem can also occur, as shown in Figure 5. While the number of samples is severely limited by the computational expense, especially for complicate engineering models, overly large sample sizes should be avoided. Further, more sample points do not necessarily lead to higher accuracy when sample size is already sufficiently large, as shown in Figure 6.

In Figures 5 and 6, results of surrogate modelling for a one-variable function with four different sample sizes are shown. We can see the impact of a proper sample size to the accuracy and efficiency of the approximation task. For higher dimensional problems and problems with more complicated input–output relations, the proper sample size is even more difficult to determine. Herein lies the problem with SSBDO method: *a priori* information regarding sufficient sample size and required accuracy of surrogate model is not available.

4.2. *Adaptive surrogate based design optimization*

To cope with the above-mentioned problems of SSBDO, an adaptive SBDO (ASBDO) approach is used instead. The framework is shown in Figure 7.

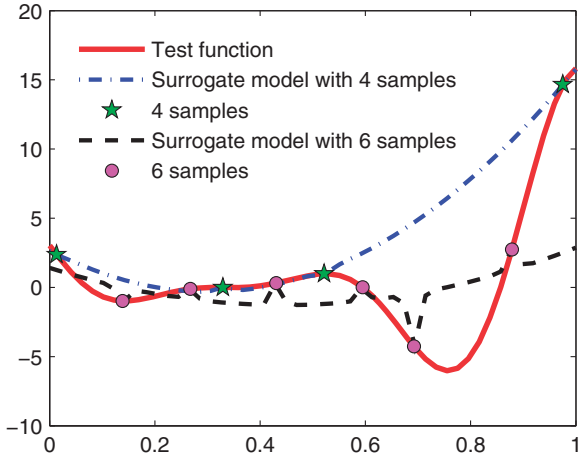

Figure 5. Surrogate modelling for a one-variable function with insufficient sample size.

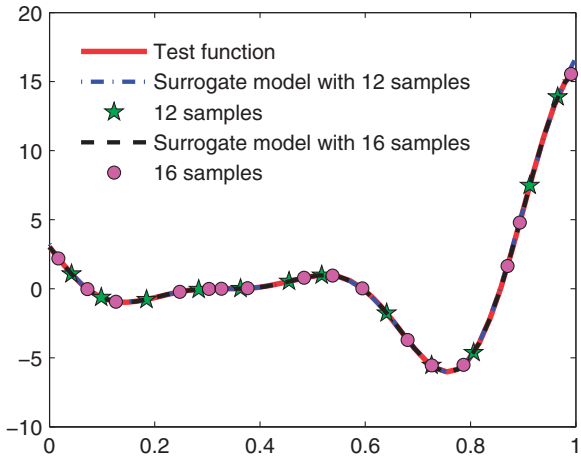

Figure 6. Surrogate modelling for a one-variable function with large sample size.

With ASBDO, we can start with a relatively small sample size to build the initial surrogate model. This model is quite simple, so that we can perform the optimization easily on this surrogate model, either with GA or with SQP. GA and SQP are two of the most typical optimization algorithms, the former method is heuristic, whereas the latter is gradient based. GA is efficient on complex multimodal optimization, and SQP is valid and robust in solving constrained nonlinear optimization problems. Because of these features, they are used here as a benchmark to test the proposed ASBDO strategy, to see how well it can support the process of optimization. After optimization is completed, the optimal results are checked, the surrogate models are validated and all such information is integrated to generate additional sample points. In this way, accuracy of the surrogate model is expected to be

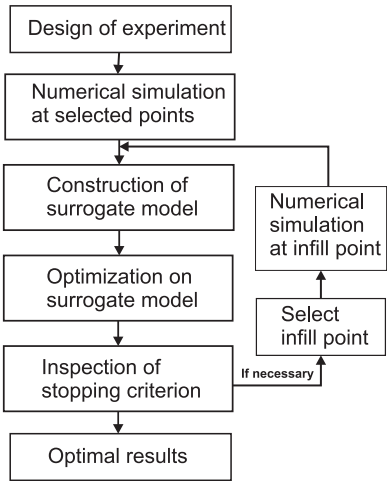

Figure 7. Framework of ASBDO.

gradually increasing, and optimization results are guided to move towards the true optimal value iteration by iteration. As the additional samples are selected according to the validation results of the optimization process, we can end up with more economical sample size. This leads to the advantage that fewer evaluations of the system simulations are needed, so that better efficiency can be acquired.

4.2.1. *Infill criterion*

Infill criterion describes how additional sample points are selected to support the progress of surrogate modelling and optimization. Surrogate models have the potential to get higher accuracy by using further infill sample points, in addition to the initial sample plan. The strategy to decide where to locate infill points is called infill criterion.

In this work, EI and CEI are used as the infill criteria. The former is used in unconstrained optimization problems and the latter is for constrained optimization. As nearly all engineering design optimization tasks are constrained, the latter criterion is used, which is developed from the former. We introduce both in the following sections.

4.2.1.1. *Expected improvement*. The idea of EI can be described as to improve the surrogate model iteratively, we have to find out where our current surrogate model is not accurate enough, or where we should give more samples to ensure the accuracy of the optimization result. To calculate the error of surrogate model, we may need extra evaluations of the true function, which is the most computationally expensive part in engineering optimization, and thus should be prevented. The surrogate models based on Gaussian process, such as Kriging, permit the calculation of an estimated error of the prediction. The estimated error at any sampled and not sampled points are calculated on the surrogate model instead of on the original

system model. This way, we avoid the expensive system model simulation. Although the obtained information is an estimated error, it guides improvement of the model, as more and more such information is gathered. We can use this advantage to locate infill points where our uncertainty in the predictions is very high.

EI is the amount of improvement we expect that the infill point will lead to the best observed design so far. We can search for the point with maximum EI, and select it as the infill point to support the surrogate model.

To get EI, we need to compute the estimated mean square error of the surrogate model first. This can be found by calculating

$$\hat{s}^2(\boldsymbol{x}) = \sigma^2\left[1 - \psi^T\boldsymbol{\Psi}^{-1}\psi + \frac{1 - \mathbf{1}^T\boldsymbol{\Psi}^{-1}\psi}{\mathbf{1}^T\boldsymbol{\Psi}^{-1}\mathbf{1}}\right], \tag{11}$$

where $\hat{s}^2(\boldsymbol{x})$ [5] is the estimated mean square error of the surrogate model at place $\boldsymbol{x}$. The other symbols are defined in Section 2.2.

EI [5] is then formulated as below:

$$E[\boldsymbol{I}(\boldsymbol{x})] = \left\{ \begin{array}{cc} (y_{\min} - \hat{y}(\boldsymbol{x}))\left(\dfrac{y_{\min} - \hat{y}(\boldsymbol{x})}{\hat{s}(\boldsymbol{x})}\right) + \hat{s}\phi\left(\dfrac{y_{\min} - \hat{y}(\boldsymbol{x})}{\hat{s}(\boldsymbol{x})}\right), & \text{if } \hat{s} > 0 \\ 0, & \text{if } \hat{s} = 0 \end{array} \right\}, \tag{12}$$

where $y_{\min}$ is the currently observed minimal, supposing that we are minimizing the objective function, $\hat{y}(\boldsymbol{x})$ is the estimated function value at place $\boldsymbol{x}$. $\boldsymbol{\Phi}$ and ϕ are the cumulative distribution function and probability density function, respectively.

The term $(y_{\min} - \hat{y})$ in EI function represents the amount of possible improvement, which is responsible for local accuracy of the surrogate model. The term $\hat{s}$ is related to the unexplored regions or uncertainty of the surrogate model, and is helpful for global exploration. By using EI, points with low objective function value and high uncertainty are given precedence, thus balanced local and global exploration can be reached. In Section 5.1, the procedure of the infill points moving from local minima to the global minima is shown with figures.

4.2.1.2. *Constrained expected improvement*. For constrained optimization problems, we have to find the infill points that not only lead to EI, but also located in, or near the boundary of the feasible area. Points in these areas are important for the constrained optimization problem and are useful to improve the accuracy of the optimization result.

An infill criterion based on CEI [5] is used, which is formulated as

$$CE[\boldsymbol{I}(\boldsymbol{x})] = E[\boldsymbol{I}(\boldsymbol{x}) \cap \boldsymbol{F}(\boldsymbol{x})], \tag{13}$$

$$E[\boldsymbol{I}(\boldsymbol{x}) \cap \boldsymbol{F}(\boldsymbol{x})] = E[\boldsymbol{I}(\boldsymbol{x})]\boldsymbol{P}[\boldsymbol{F}(\boldsymbol{x})], \tag{14}$$

where $\mathbf{CE}[\boldsymbol{I}(\boldsymbol{x})]$ is the CEI at $\boldsymbol{x}$. $\boldsymbol{I}(\boldsymbol{x}) \cap \boldsymbol{F}(\boldsymbol{x})$ is the intersection of the set of $\boldsymbol{x}$ that will lead to certain improvement and are located in the estimated feasible area. $\mathbf{E}[\boldsymbol{I}(\boldsymbol{x}) \cap \boldsymbol{F}(\boldsymbol{x})]$ is the expected value of improvement at $\boldsymbol{x}$ of the intersection. $\boldsymbol{P}[\boldsymbol{F}(\boldsymbol{x})]$ is

the probability that $\boldsymbol{x}$ is located in the feasible area, and can be calculated as follows:

$$\boldsymbol{P}[\boldsymbol{F}(\boldsymbol{x})] = \frac{1}{\hat{\boldsymbol{s}}\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{[F-\hat{g}(x)]^2}{2s^2}} \mathrm{d}\boldsymbol{G}, \tag{15}$$

where $\boldsymbol{F}$ is the measure of feasibility ($\boldsymbol{G} - g_{\min}$), $\boldsymbol{G}(\boldsymbol{x})$ is a random variable, g is the constraint function and $g_{\min}$ is the boundary value of the constraint [5]. As the surrogate model is improving gradually, the estimated feasible boundary is approaching the true feasible boundary, so as the infill points. By using CEI, points with low objective function (for minimization) value and high uncertainty near the feasible area are given precedence.

Having the formulation of CEI, we are able to use it as infill criterion to support the SBDO. Point with maximum CEI is searched in each iteration, and the system response is evaluated there to support a new iteration.

In Section 4.2.3, we implement the infill criteria into the framework of ASBDO. In Section 5, ASBDO with EI or CEI as infill criteria is used in the numerical tests of non-constrained and constrained optimization problems, respectively.

4.2.2. *Stopping criterion*

Stopping criterion are also very important for an algorithm to successfully converge to an acceptable optimum, especially for ASBDO, which is developed to reduce the number of model evaluations. To set up a proper stopping criterion for ASBDO, there are two risks to be aware of:

(1) We should not stop the iteration when the surrogate model has small change between two iterations. There is situation when EI or CEI is in the local exploitation stage, so that infill points are very close to each other in several continuous iterations. In such situation, we cannot ensure that one infill point in iteration will lead large enough improvement to the surrogate model, thus it will be unwise to stop there.
(2) We should not stop the iteration when the optimal results we find vary quite little between two iterations. There is possibility that some areas of great uncertainty have not yet been discovered, and it is wiser to check other validation information first.

A good sign that our optimal result $\boldsymbol{x}^*$ is close to the true optimum is that the value of the constraint function at $\boldsymbol{x}^*$ is very close to its boundary value. Therefore, we can always check the constraint function value when we get an optimal result in each iteration. Thus the stopping criteria used in this study are as follows:

(1) Distance of the infill points found between two iterations is smaller than a predefined value δ_1.
(2) Distance of the optimal results obtained in two iterations is smaller than a predefined value δ_2.
(3) Distance between the constraint function value at current optimum $\boldsymbol{x}^*$ and the constraint boundary is smaller than a predefined value δ_3.
(4) Iteration number reaches the maximum allowable value $n_{\max}$.

The algorithm is stopped when: (1) and (3) are satisfied; (2) and (3) are satisfied or (4) is satisfied.

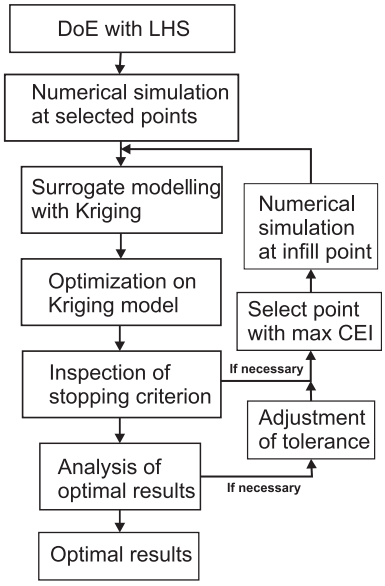

Figure 8. Framework of the detailed procedure of ASBDO implemented in this work.

4.2.3. *Implementation of ASBDO*

The detailed procedure of ASBDO proposed in this article is shown in Figure 8.

Initial samples are generated using LHS. The initial number of samples can be relatively small, but should not be too small considering the balance of accuracy and time expense. If the distribution of samples in the design space is too sparse, the time needed for adaptive sampling will increase for required accuracy. Although we can ensure smaller evaluation number than using DO and SBDO methods, a proper initial sample size is still not known *a priori*, which will be studied in future work. After numerical simulation of our original model at the selected initial sample points, a surrogate model is then built using Kriging model.

Optimization is performed on the surrogate model instead of on the original model with optimization algorithms, such as SQP or GA. As we know, SQP typically needs fewer evaluations than GA, but has trouble when either the objective function or the constraint function of the optimization problem is multimodal, noisy or non-convex. The position of the initial design point is very important for SQP, here the initial design point is always selected from the sample points that has the best objective value.

The stopping criterion is checked before a new iteration begins. If it is satisfied, the optimal results are analysed. The objective value and constraint value at the optimal point should be validated, the Karush–Kuhn–Tucker condition should be fulfilled when using SQP, the active constraints should be checked, etc. If necessary, adjustment is made to the predefined values or the tolerance value in the stopping criteria, and a new iteration is started.

If the stopping criteria are not satisfied, the iteration goes on to find infill points. An optimization procedure is done for the EI or CEI function. A point with

maximum (C)EI value is searched and selected as the infill point. Because the infill criterion function is not expensive to compute, but is often multimodal, GA is used to find the maximum. Then numerical simulation is done at the selected point, and this information is used to rebuild a surrogate model and to run next iteration. The ASBDO process continues until the stopping criterion is satisfied and the analysis of the optimal results is passed.

5. Numerical tests of the ASBDO strategy

In this section, three numerical tests are completed with the proposed ASBDO strategy. In the examples of a one-variable function, we can see intuitively how ASBDO works iteration by iteration. This makes it easier to find problems and solutions for improvements. The second example is a two-dimensional function that we can still show the results visually. The last example is a three-dimensional airplane wing model, which is solved using FEM, so that we can see how much computational efforts our method can save for a simple engineering problem.

5.1. *Tests on a one-variable function*

The object of the first example is to get the minimal value of the one-variable test function [5]:

$$\min f(x) = (6x-2)^2 \sin(12x-4), \quad x \in [0,1]. \tag{16}$$

This function has one local minimum at $(0.1426, -0.9863)$ and one global minimum at $(0.7572, -6.0207)$. The performance of using EI as infill criterion is shown iteration by iteration in Figures 9–14.

In Figures 9–14, the stars represent the sample points, and the square in each figure represents the next infill point, which is found in the current iteration with maximum EI. It will be added to the sample sets and will enhance surrogate model.

It can be seen from the first iteration that the surrogate model is built with three initial sample points. From the curve of EI, we find that the maximum EI happens at

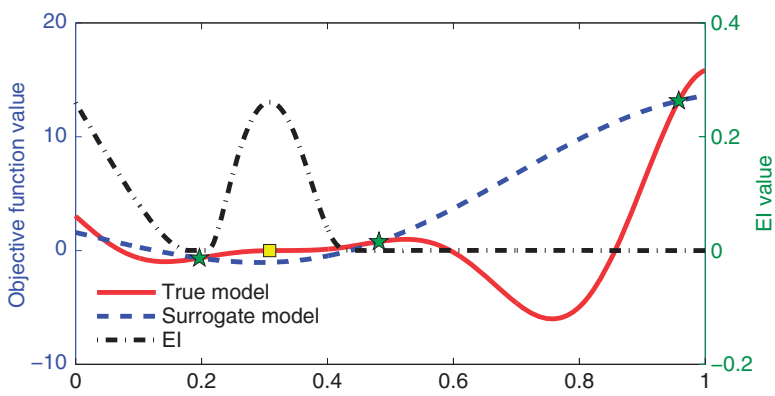

Figure 9. The first iteration of ASBDO with EI.

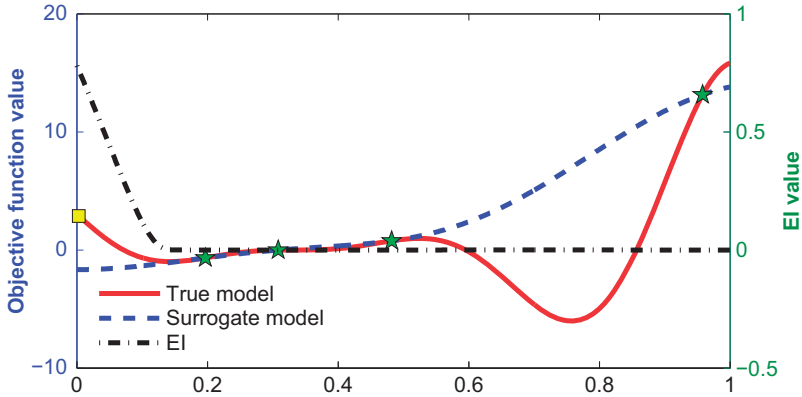

Figure 10. The second iteration of ASBDO with EI.

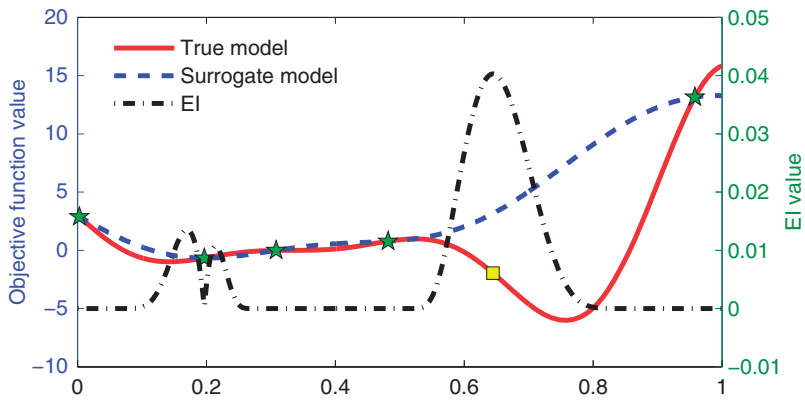

Figure 11. The third iteration of ASBDO with EI.

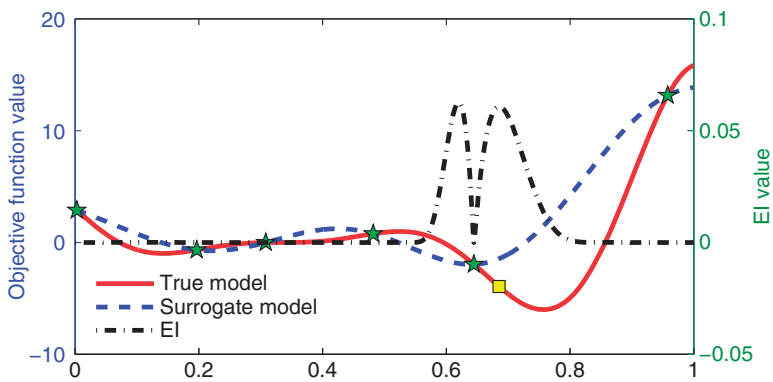

Figure 12. The fourth iteration of ASBDO with EI.

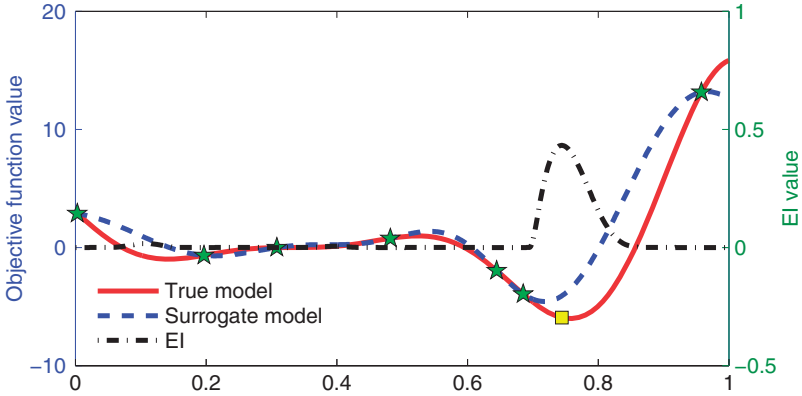

Figure 13. The fifth iteration of ASBDO with EI.

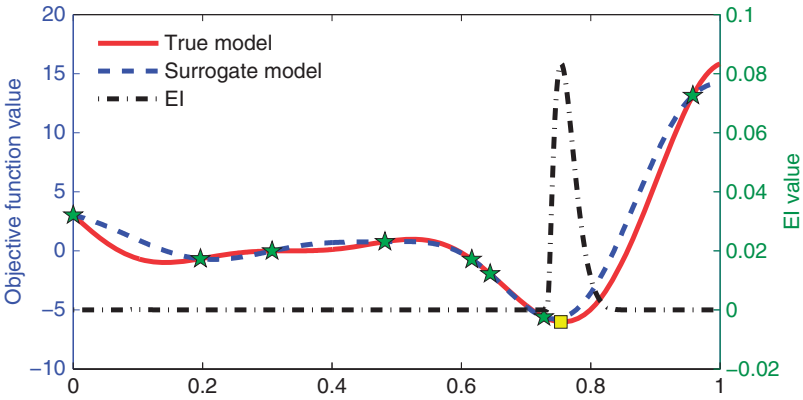

Figure 14. The sixth iteration of ASBDO with EI.

the current minimum of the surrogate model, and this point is chosen as infill point. We can see that the area of the global minimal of the true model is totally unexplored, so that EI cannot find any improvement there and the EI values are all close to zero in that area. The point that will lead the maximum improvement to the surrogate model is thus the current minimal.

We can see from the second iteration that the new surrogate model is now built with four samples, three of them are initial samples and one is the infill point. The new surrogate model is of higher accuracy, and finds a new infill point. As in the first iteration, the global optimal area is still not explored, so the infill point is still found at the place of current local minimal.

In the third iteration, we can see that after adding the second infill point, the surrogate model is very close to the true model at the left side, and the EI curve can realize that either the current minimal or the unexplored area can lead to high EI,

Table 1. SQP optimization of one-variable function with three different methods.

	DO	SSBDO	ASBDO
$\overline{n}_{eva}$	44	12	10
p_{acc}	35%	100%	100%
$rmse(x^*)$	0.4915	2.6482e−4	1.7089e−4
$rmse(f^*)$	4.0856	0.0081	0.0014

Table 2. GA optimization of one-variable function with three different methods.

	DO	SSBDO	ASBDO
$\overline{n}_{eva}$	1020	12	10
p_{acc}	100%	100%	100%
$rmse(x^*)$	0	2.0551e−4	1.0680e−4
$rmse(f^*)$	0	0.0109	0.0029

as can be seen in Figure 11. One important advantage of using EI as infill criterion is that the global optimal area, when found, will always have larger value for EI than the local optimal area. This advantage is very helpful to guide us to find the global optimum. We can see that in the following three iterations the infill points are found closer and closer to the true optimum. Finally, with nine sample points in all, we get an accurate optimization result. DO and SSBDO usually take more evaluations to get a global optimum.

To access the performance of the ABSDO, the quality of the optimization run is compared with three different optimization approaches: (1) DO, (2) SSBDO and (3) ASBDO. Considering the robustness of results, optimization with each approach is run 100 times, and the average and statistical results are used. Here in Tables 1 and 2 results of the three methods are listed. The optimization algorithms in two tables are SQP and GA, respectively. For DO, as we have no *a priori* knowledge of the objective function, the initial design point is generated randomly among the 100 runs.

Here $\overline{n}_{eva}$ is used to represent average number of evaluations of the true model, and p_{acc} is the percentage of the acceptable extreme values found among the 100 runs. $rmse(x^*)$ and $rmse(f^*)$ are the RMSE of the optimum design variable and optimum objective value of the 100 runs.

It can be seen in Tables 1 and 2 that ASBDO has the best performance compared with both SQP and GA algorithms. It needs the least evaluations of the true function, and all the solutions found in the 100 runs reached global optimum and with good accuracy.

As is known, that SQP tends to stop at a local optimum if initial design point is not properly chosen. Here the reason why SSBDO and ASBDO can always find the global optimum, even with SQP algorithm can be explained. For surrogate-based methods, the current best sample point is used as initial design point. With a good

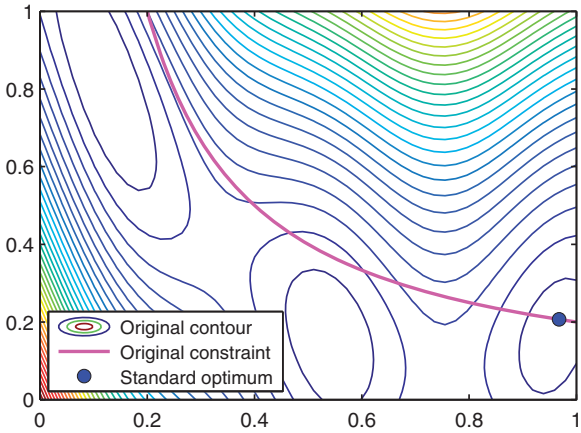

Figure 15. Original normalized contour graph and standard optimal result of constrained Branin function.

distributed sample sets and infill points, we reduce the risk of becoming trapped in the area of local minima.

5.2. *Tests on a two-variable function*

Here we minimize the two-variable Branin function [5], which is defined as follows:

$$\begin{gathered}\min f(x) = \left(x_2 - \frac{5.1}{4\pi^2}x_2 + \frac{5}{\pi}x_1 - 6\right)^2 + 10\left[\left(1 - \frac{1}{8\pi}\right)\cos x_1 + 1\right] + 5x_1 \\ \text{s.t.} \quad x_1 \in [-5, 10], \quad x_2 \in [0, 15], \\ \overline{x}_1 * \overline{x}_2 > 0.2, \\ \overline{x}_1 \in [0, 1], \quad \overline{x}_2 \in [0, 1],\end{gathered} \tag{17}$$

where $\overline{x}_1$ and $\overline{x}_2$ are normalized design variables, which satisfies the following conditions:

$$\overline{x}_i = \frac{x_i - lb_i}{ub_i - lb_i}, i = 1, 2 \tag{18}$$

where lb and ub are the lower and upper bounds of the design variables. Surrogate modelling process is done on the normalized design variables that are in the range of $[0, 1]$, and LHSs are also generated between $[0, 1]$ for each dimension. The function value is calculated on the true range, and we have to change design variables into the true range whenever necessary, with the following expression:

$$x_i = \overline{x}_i * (ub_i - lb_i) + lb_i, \quad i = 1, 2. \tag{19}$$

For this constrained optimization problem, we build surrogate models for both objective function and constraint function. The CEI introduced in Section 4.2.1 is used as infill criteria. The contour graph of the constrained Branin function and the standard optimization result is shown in Figure 15. The horizontal and vertical axes

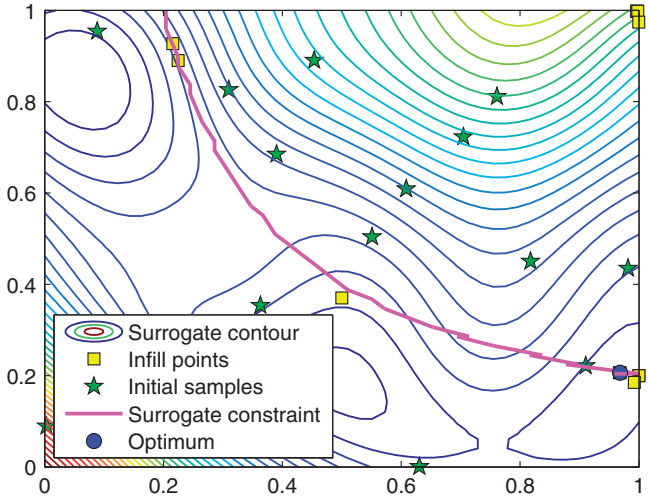

Figure 16. Surrogate normalized contour graph of constrained Branin function and the optimal result with ASBDO.

Table 3. SQP optimization of Branin function with three different methods.

	DO	SSBDO	ASBDO
$\bar{n}_{eva}$	418	100	39
p_{acc}	55%	42%	100%
$rmse(x^*)$	0.5193	0.5071	0.0161
$rmse(f^*)$	12.5320	11.6562	0.7323

represent the normalized two design variables, respectively. The line crossing the contour lines represents the constraint boarder. It can be seen that there are two local minima and one global minimum for the Branin function. For the constrained Branin function, the only minimum is located at the right-hand side bottom on the constrained boarder. The optimization result using ASBDO approach is demonstrated in Figure 16.

Figure 16 shows that ASBDO can find the right minimum with few samples and infill points. We can see that with CEI infill criterion, the infill points are all located near the constrained boarder or the boundary of the design variable, and end up near the standard minimum point. Infill points here help improve the surrogate model and lead to the right optimum efficiently. Although both the objective function and the constrained function are surrogate models, we get good performance in optimization with ASBDO. To compare the quality of three different approaches, DO, SSBDO and ASBDO, the optimal results are shown in Tables 3 and 4. As in the first test, all three processes are run 100 times, using SQP and GA as optimization algorithm.

Table 4. GA optimization of Branin function with three different methods.

	DO	SSBDO	ASBDO
$\bar{n}_{eva}$	9024	100	39
p_{acc}	94%	96%	100%
$rmse(x^*)$	0.2140	0.1483	0.0210
$rmse(f^*)$	5.2536	3.4839	0.8324

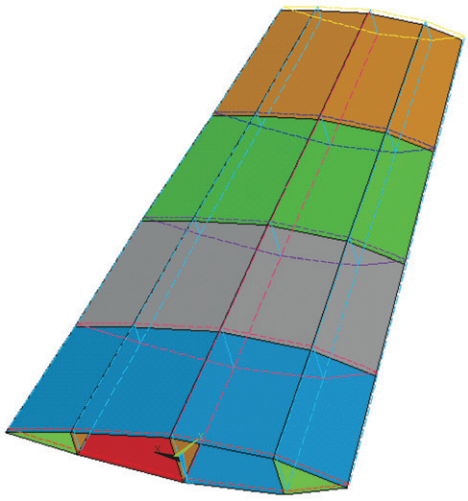

Figure 17. A simplified aircraft wing model.

It can be seen that ASBDO has better performance than DO and SSBDO, with a great reduction of evaluations, much larger chance of obtaining the optimum and much higher accuracy, either with SQP or with GA. The performance of DO is restrained by the choice of the initial design point. Although it has accurate system models, a global optimum can only be found when the initial design point is given at a proper region, which is not known *a priori*. The SSBDO method is highly dependent on the DoEs. The quality of the surrogate system models can be poor because a DoE may lack information in the neighbourhood of the global optimum. SSBDO cannot refine the DoE and therefore the possibility of getting the optimal value is also limited. The reason why ASBDO has higher chance of finding right optimum value can be attributed to the optimization strategy and the infill criterion. The CEI criterion has the potential to find the global optimum, as the CEI value is larger in the area of global optimum than in the area of the local optimum.

5.3. *Tests on a finite element model of an aircraft wing*

A simplified aircraft wing model is used here to test the performance of ASBDO method on FEM models and on higher dimensional problems. For the aircraft wing model shown in Figure 17, there are 15 design variables: geometrical parameters,

Table 5. SQP optimization of a 3D aircraft wing model with two different methods.

	DO	ASBDO
n_{Ite}	16	19
n_{eva}(FEM)	291	155
n_{eva}(SgM)	–	18002
Time[s]	2500.7	1693.8
g^*	2.985e−7	−9.405e−5
f^*	14.4445	14.6157

including thickness of skin, ribs, spars, as well as the cross-section area of the spars. The objective function is the total mass of the wing. The boundary condition is simplified so that the wing is fixed on the root surface. The load condition analysed here is static pressure distributed on the surface of the skin. The optimization problem can be expressed by

$$\begin{aligned} \min \quad & f(\boldsymbol{x}) = \boldsymbol{m} \\ \text{s.t.} \quad & g(\boldsymbol{x}) = \frac{\sigma_{\max}}{\sigma_{allow}} - 1 \leq 0, \\ & \boldsymbol{x} = [x_1, x_2, \ldots, x_{15}], \\ & lb_i \leq x_i \leq ub_i, \\ & i = 1, 2, \ldots, x_{15}, \end{aligned} \tag{20}$$

where $\boldsymbol{m}$ is the mass function, σ is the stress generated on the wing and σ_{allow} is the allowable stress. lb and ub are the lower and upper bounds of the design variables.

The stress of the wing is obtained using an FEM solver, which is more computational costly than the mathematical models tested in Sections 5.1 and 5.2. Here the optimization is completed using SQP algorithm for both the DO and the ASBDO approaches. The iteration results of these two approaches can be seen in Table 5.

In this table, n_{Ite} is defined as the number of iterations, which has different meanings for DO and ASBDO. The DO approach takes 16 iterations to finish the task, each with a number of evaluations of the FEM model. The ASBDO approach takes 19 iterations of infilling process, each with only one evaluation of the FEM model plus a few evaluations of the surrogate models. As in this model one evaluation of the surrogate model takes only about 1/500 of the time needed by an FEM model, time expense can be greatly reduced.

n_{eva}(FEM) and n_{eva}(SgM) are the numbers of evaluations of the FEM model and the surrogate model, respectively. It can be seen from Table 5 that with ASBDO, we can save about 50% evaluations of the FEM model, which is the most time consuming part of the optimization process. Although we need 18,002 evaluations of surrogate model by using ASBDO strategy, as has been explained, the time expense for each evaluation is quite small compared with that of the FEM model. As a result, ASBDO can reduce 32.3% of the time expense needed for the DO approach.

Besides those two measures, we define g^* as the final maximum constraint value, which is used as an additional measure of accuracy. It indicates whether an algorithm has converged at the right optimum. It should be a negative number so that the optimal result is feasible, and it should be close to zero, so that we have reached an

active constraint and the optimal result is more likely to be an extreme. f^* is also defined here as the final optimal objective function value to measure accuracy and feasibility. Judging from the measures of g^* and f^*, we can see that the ASBDO method managed to find a good feasible design.

6. Conclusion

To reduce computational expense in structural design optimization problems, surrogate models are used to approximate the time-consuming FEM models. Traditional SBDO method has the major shortcoming that proper sample size is not known *a priori*. In this study, an ASBDO approach is proposed. It can start with very small sample size and then select additional sample points according to the surrogate based optimal results. The infill criteria: EI and CEI have the advantage that information from the optimization process is sufficiently utilized to guide the selection of additional samples. As a result, we can solve the optimization problem with higher efficiency and good accuracy.

It is expected that the ASBDO approach can be conveniently adapted to larger and more complicated design optimization problems and get more benefits in saving computational and time expense. However, the curse of dimensionality is still a problem to be further studied for this method, although the infill strategy helps to identify some of the local optima in a multimodal function.

Future work will include the following aspects:

(1) Study of the determination of parameters to be surrogated: As discussed in Section 3, instead of approximating the design variables directly, we can study the relationship between inputs and outputs, and approximate a variation form of the design variables, which is easier to make closer approximation. This helps to further increase the accuracy and efficiency of surrogate modelling approach.
(2) Study of the infill criteria and multiple infill strategy: It is expected that with the infill criteria defined to find multiple infill points or the selection strategy to choose all the necessary support points, we may better handle the curse of dimensionality, facilitate parallel computation and further increase efficiency.
(3) Study of the rules of setting initial sample size in ASBDO method: Although we have already greatly reduced the sample size using ASBDO, the initial sample size still has a large affect to the accuracy and efficiency of surrogate modelling process. Guidelines to set initial sample size may help improve the robustness of our method.
(4) Stopping criterion: Stopping criterion are vital and need to be refined so as to further reduce the time expense of ASBDO and improve the quality of optimal results.
(5) Use of gradient information: Gradient information of surrogate models can be considered, which may potentially reduce time expense and result in better accuracy.
(6) Tests on larger computational expensive problems: Tests on more complex engineering problems should be done to increase the general applicability of this ASBDO method.

Acknowledgements

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